

Name:	Tutor group:
Teacher:	Science set:

**Pearson Edexcel IGCSE
Double Award Science
Island School Chemistry UNIT 1**

Matter and Separation Techniques

(Y10 2026-2027)

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Lesson 1 - States of Matter

Lesson 2 - Diffusion and Dissolving

Lesson 3 - Elements, Compounds and Mixtures

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Lesson 5 - Distillation

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Revision worksheets

UNIT 1 Syllabus Statements

Lesson	Syllabus statement from the exam board	Textbook pages
1 States of Matter	1.1 understand the three states of matter in terms of the arrangement, movement and energy of the particles. 1.2 understand the interconversions between the three states of matter in terms of: • the names of the interconversions • how they are achieved • the changes in arrangement, movement and energy of the particles.	Read pages 6 - 8 Study Questions on page 8
2 Diffusion and Dissolving	1.3 understand how the results of experiments involving the dilution of coloured solutions and diffusion of gases can be explained. 1.4 know what is meant by the terms: • solvent • solute • solution • saturated solution.	Read pages 2 - 5 Study Questions on page 5
3 Elements, Compounds and Mixtures	1.8 understand how to classify a substance as an element, compound or mixture 1.9 understand that a pure substance has a fixed melting and boiling point, but that a mixture may melt or boil over a range of temperatures. 1.14 know what is meant by the terms atom and molecule	Read pages 9 - 11 Study Questions on page 11
4 Filtration and Crystallisation	1.14 know what is meant by the terms atom and molecule 1.10 describe these experimental techniques for the separation of mixtures: • filtration • crystallisation	Read pages 12 - 17 Study Questions on page 17
5 Distillation	1.10 describe these experimental techniques for the separation of mixtures: • simple distillation • fractional distillation	
6 Chromatography	1.10 describe these experimental techniques for the separation of mixtures: • paper chromatography. 1.11 understand how a chromatogram provides information about the composition of a mixture 1.12 understand how to use the calculation of R _f values to identify the components of a mixture 1.13 practical: investigate paper chromatography using inks/food colourings	

Filtration separates an insoluble solid from a liquid: the solid remains as the residue in the filter paper and the liquid passes through as the filtrate.

Fractional distillation separates miscible liquids with different boiling points (e.g. ethanol and water) using a fractionating column; the lowest boiling liquid distils first.

Reversible changes of state: melting (solid-liquid) and boiling/evaporation (liquid-gas) on heating; freezing and condensation on cooling; sublimation (solid-gas).

Gas particles move randomly and spread (diffusion). Lighter particles move faster, so ammonia (Mr 17) diffuses faster than hydrogen chloride (Mr 36.5) - ring forms nearer the HCl end.

Unit 1 Glossary

By the end of this unit, you should be familiar with the following terms. You will need to correctly use these words during any assessments.

Atom	The smallest part of an element that can exist.
Particles	Extremely small pieces of matter, for example, atoms, molecules and ions.
Molecule	A molecule consists of an exact number of atoms joined together covalently.
Element	A pure substance that cannot be split up into anything simpler. An element contains just one type of atom.
Symbols	Chemists use symbols to represent atoms. Every element has a different symbol. For example, the symbol for carbon is C.
Diffusion	The random movement and mixing of particles in a liquid or gas.
Dissolving	Solid particles break apart and exist in the gaps between the liquid particles.
Simple distillation	A separation technique that allows a pure liquid to be separated from a mixture. Involves the liquid boiling and then condensing.
Filtration	A separation technique that allows an insoluble solid to be separated from a mixture in a liquid. The liquid passes through the filter paper but the solid does not.
Crystallisation	A separation technique that allows a soluble solid to be separated from a dissolved mixture in a liquid. The mixture is heated and the liquid partially evaporates. As the mixture cools, the solid starts to appear. This is called crystallisation.
Paper chromatography	A technique that allows a complex mixture of substances to be separated and identified. Each substance rises up the filter paper by different distances.
Retention factor (R_f value)	Used to identify the substances used in paper chromatography. The distance moved by the solute / distance moved by the solvent.
Solute	Solid particles that you are trying to dissolve.
Solvent	Liquid that you are using to dissolve the solid. Water is the most common solvent, but there are others such as propanone is used to dissolve nail polish.
Solution	When the solid has dissolved in the liquid, we call this a solution.
Saturated solution	This is the maximum amount of solute that a solvent can dissolve at a given temperature.
Solubility	This is a measure of how much solute has been dissolved in the solvent. It is often measured in g of solute per 100g of solvent at a specific temperature)

States of Matter

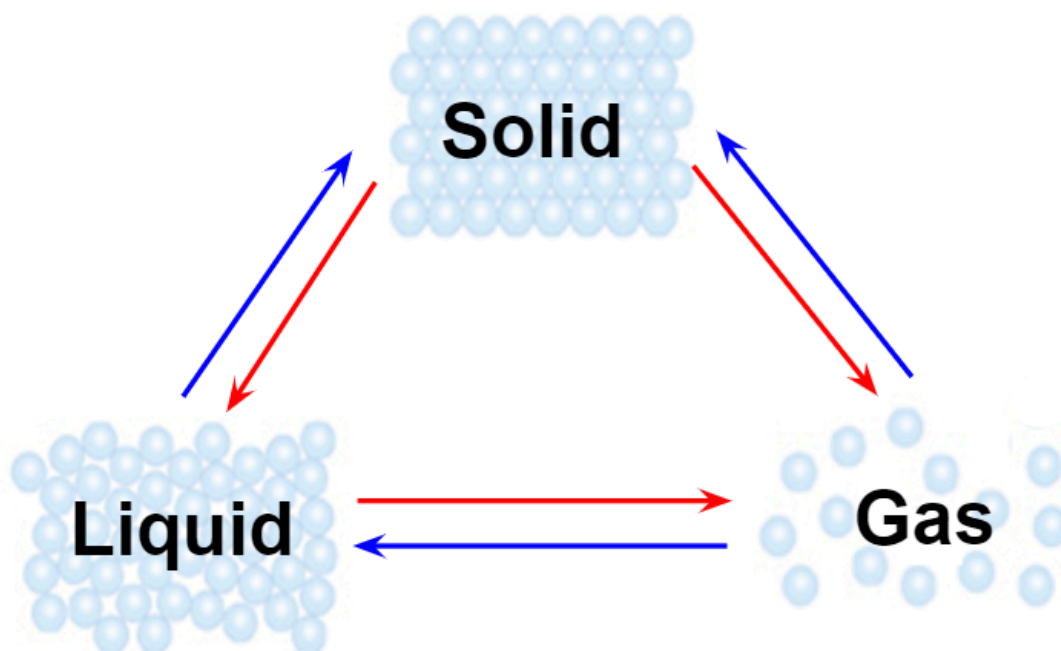
1. Complete the following table to summarise the properties of solids, liquids and gases (some have been done for you)

Solid: fixed shape + volume, particles close in rows, vibrate only. Liquid: fixed volume, takes container shape, particles slide. Gas: fills container, particles far apart, fast random motion

State	Solid	Liquid	Gas
Arrangement	Very close together Regular arrangement	Close together with a random arrangement	Far apart with a random arrangement
Motion	Vibrate about fixed positions	Slip and slide over each other	Move quickly and randomly in all directions
Properties (shape, volume & compressibility)	Definite shape Not compressible	Definite volume; takes the shape of the container; not compressible Definite volume	Spread out to fill available volume Volume depends on the container Compressible
Model of particles	Draw the particles regularly arranged and close together (touching)	Draw the particles close together but randomly arranged (touching) 	Draw a few particles far apart and randomly arranged

2. a) The particles in which of the 3 states of matter have the most energy? **gas**.....
 b) In which states of matter are particles touching? **solid and liquid** and
 c) The particles in which of the 3 states of matter are furthest apart? **gas**.....

3. Solids, liquids and gases can be interconverted by either heating or cooling. Complete the following diagram by adding the names for the interconversions on the appropriate arrows.



· Label the arrows: solid to liquid = melting; liquid to solid = freezing; liquid to gas = boiling (evaporating); gas to liquid = condensing; solid to gas = sublimation; gas to solid = deposition.

4. What is the difference between evaporation and boiling?
 Evaporation is the surface phenomenon and evaporation occurs throughout the liquid at one fixed temperature (the boiling point).

5. Convert the following statements to chemical equations:

a) Water turning into steam



b) Iron (Fe) melting



c) Dry ice turning into carbon dioxide gas

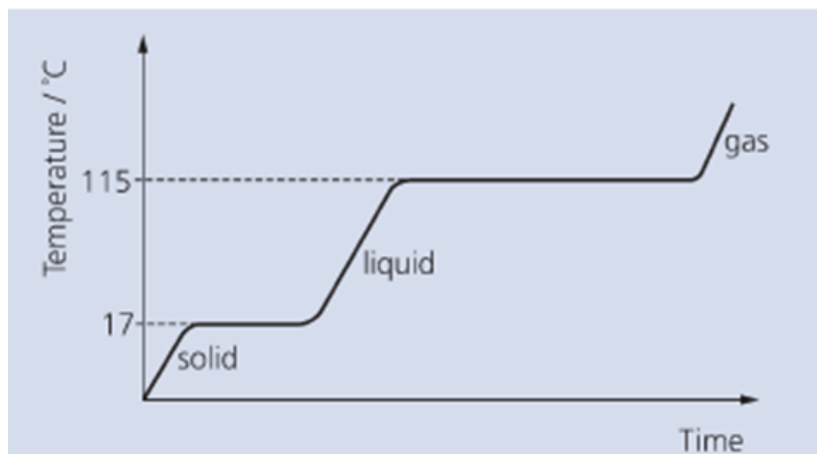


d) Liquifying oxygen from the air



States of Matter

1. Below is a heating curve for a pure substance. It shows how the temperature rises over time, when the substance is heated until it melts, then boils.



a) What is the melting point of the substance? *Read the temperature of the first flat plateau on the curve - that is the melting point.*

b) What happens to the temperature while the substance changes state?

.....

..... *The temperature stays the same (constant), because the energy supplied is used to overcome the forces between the particles, not to raise the temperature.*

c) The graph shows that the substance takes longer to boil than to melt. Suggest a reason for this.

.....

.....

..... *Boiling needs more energy than melting: it completely overcomes the forces between all the particles so they become a gas; melting only loosens the regular arrangement.*

d) How can you tell that the substance is not water?

e) What is the state of this substance at room temperature (25°C)?

f) Carefully sketch a cooling curve for this substance

Water melts at 0 °C and boils at 100 °C, but this substance has different melting and boiling points.

Liquid - 25 °C lies between its melting point and its boiling point.

· Cooling curve: temperature falls, then a flat plateau at the boiling point, falls again, then a second flat plateau at the melting point.

g) Describe the *change* in **arrangement**, **movement** and **energy** of the particles as they change state from solid to liquid.

Arrangement: ~~Particles move further apart and the regular arrangement becomes a random one (particles are no longer in fixed positions).~~

.....

.....

Movement: ~~From vibrating about fixed positions to moving randomly, sliding and slipping over each other.~~

.....

.....

Energy: ~~The particles gain (kinetic) energy, so they move faster.~~

.....

.....

h) Using information from the first table below, complete the second table with the words 'solid', 'liquid' or 'gas'

~~Particles closely packed in regular rows · solid; particles close together but~~

Substance	Melting Point (°C)	Boiling point (°C)
randomly arranged · liquid; particles far apart and randomly arranged · gas.		
water	0	100
methane	-182	-162
oxygen	-219	-183
ethanol	-117	79

Substance	State at 25°C	State at 1000°C	State at -50°C
water	water: 25 C = liquid, 1000 C = gas, -50 C = solid		
methane	methane: 25 C = gas, 1000 C = gas, -50 C = gas		
oxygen	oxygen: 25 C = gas, 1000 C = gas, -50 C = gas		
ethanol	ethanol: 25 C = liquid, 1000 C = gas, -50 C = liquid		

Dissolving & Diffusion

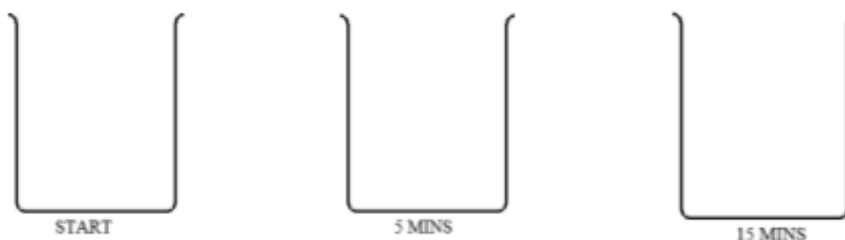
Method: Carefully put a purple crystal of potassium permanganate (**solute**) into a beaker of water (**solvent**) using a hollow glass tube. Do not move the beaker but check it after 5 and 15 minutes

Draw three particle diagram below to explain what is happening to the **solution over time**.

· Draw 3 particle diagrams: (1) purple crystal on the bottom with the water particles; (2) purple particles beginning to spread up through the water; (3) purple particles evenly spread through the water.



= 1 crystal of potassium permanganate



2. Explain what is happening to the particles in your diagram. Try to use as many of the following words as possible.

The purple crystals are the solute. They dissolve in the water (solvent) to form a solution. The solute particles move randomly and spread between the water particles by diffusion.

solvent	diffusion	move	particles	solution	dissolves	spreads	water	solute
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3. How could you speed up the processes taking place in the beaker in the experiment above? Explain why this would speed it up in terms of the **energy** of the particles involved.

Stir the mixture, warm it, or use smaller crystals of potassium permanganate (more surface area).

Heating gives the particles more kinetic energy, so they move faster and spread out and mix more quickly.

4 Label the following gas jars after the experiment after the cover slip is removed:



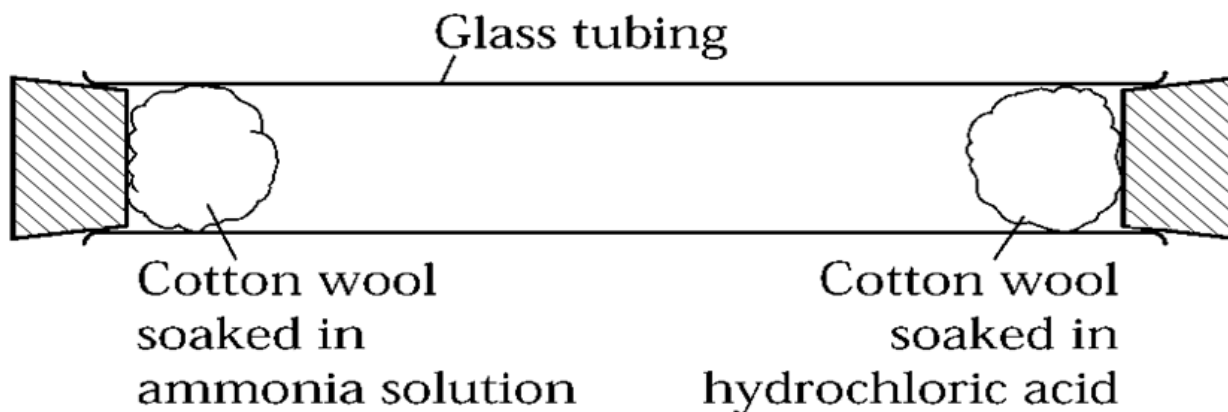
· Label the gas jars: the brown vapour has spread evenly through both jars because the gas particles diffuse to fill the space.

5 Diffusion happens faster in gases than in liquids. Explain why this happens.

Gas particles are much further apart with (almost) no forces between them, so they move faster and mix more quickly than liquid particles.

Diffusion of Gases

In this experiment (demo) you will investigate what happens when two different gases diffuse through the air. Hydrogen chloride and ammonia will react together to give a white solid. When the two gases meet, this appears as white smoke in the air. Particles of hydrogen chloride are about twice as heavy as particles of ammonia.



Recording your results

1. When your teacher puts the cotton wool in the tube, look carefully to see where the white smoke ring appears. *Mark this clearly on the diagram.*

Considering your results/conclusions

2. Which gas moved faster, ammonia or hydrogen chloride? **ammonia**.....
3. Explain how you can tell this from the results

The white ring forms nearer the hydrogen chloride end, so the ammonia particles must have travelled further in the same time - ammonia diffused faster.

4. What would happen to the time taken to form the white smoke ring if the tube was heated? And where would the ring form? Explain why.

.....

The ring would form in a shorter time and in roughly the same place, because heating gives all the particles more kinetic energy so both gases move faster.

5. *On the above diagram* draw a line to illustrate the path taken by an ammonia molecule and hydrogen chloride molecule as they move along the tube

6. The particles in a gas move at a very fast speed. Explain why it took some time for the white smoke ring to form.

.....

*The gas particles move randomly and collide with the air particles, so they take time to travel along the tube and meet.
 • Mark the white smoke ring nearer to the hydrogen chloride end of the tube (about one third of the way along from the HCl end).*

Extension: Give an equation to show the reaction between HCl and NH₃ to give a white smoke.

Given that $KE = \frac{1}{2}mv^2$, explain why gases with a smaller mass tend to diffuse faster at the same temperature. Use the internet to research and explain this.

• Draw a jagged/random path from each end of the tube: a longer path from the ammonia end and a shorter path from the hydrogen chloride end, meeting at the ring.



Elements, Compounds & Mixtures

Element names and symbols:

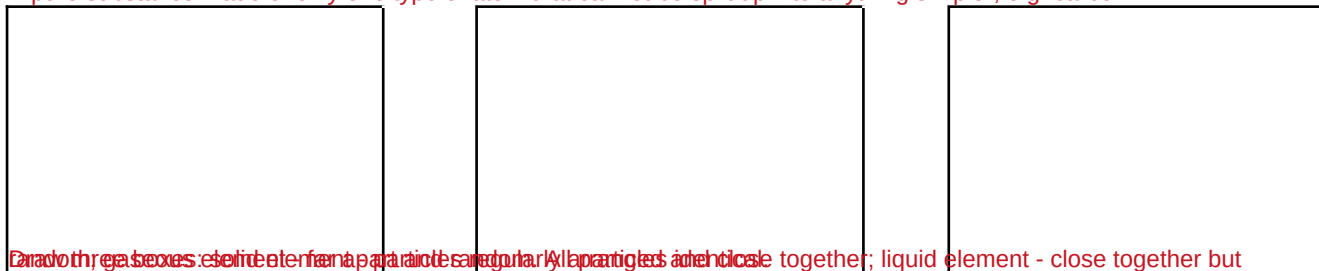
- | | | |
|----|-----|----------------|
| 1) | 6) | 6) carbon, C |
| 2) | 7) | 7) nitrogen, N |
| 3) | 8) | 8) oxygen, O |
| 4) | 9) | 9) fluorine, F |
| 5) | 10) | 10) neon, Ne |
- 1) hydrogen, H
2) helium, He

Define the term **element**:

- 3) lithium, Li
4) beryllium, Be
5) boron, B

Particle diagrams for a solid, liquid and gaseous element:

A pure substance made of only one type of atom that cannot be split up into anything simpler, e.g. carbon.



Can draw three boxes: solid element - regular arrangement of particles; liquid element - close together but

Define the term **compound**:

.....

What does the formula H_2O mean?

A pure substance made of two or more different elements chemically joined (bonded) together, e.g. H_2O .

It shows a water molecule is made of 2 hydrogen atoms chemically joined to 1 oxygen atom.

Define the term **molecule**: A molecule is two or more atoms joined together by chemical bonds, e.g. O_2 or H_2O .

.....

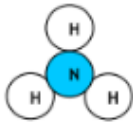
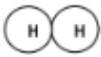
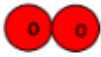
Particle diagrams for H₂O in the solid, liquid and gaseous states:

Draw water particles close together in a regular arrangement (touching).

Draw water particles close together but randomly arranged (touching).

Draw water particles far apart and randomly arranged.

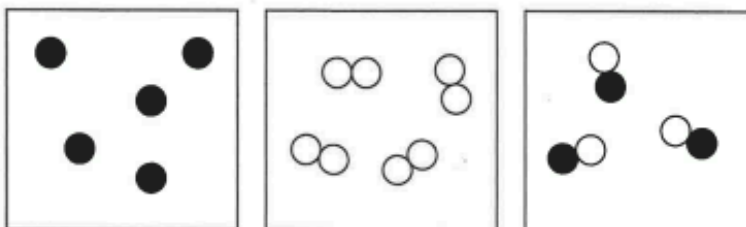
Fill in the blanks

Name of substance	Diagram of molecule	Chemical formula	Element or compound?
Water			
Nitrogen			
Carbon dioxide			
Ammonia			
Sulfur dioxide			
Hydrogen			
Hydrogen chloride			
Oxygen			

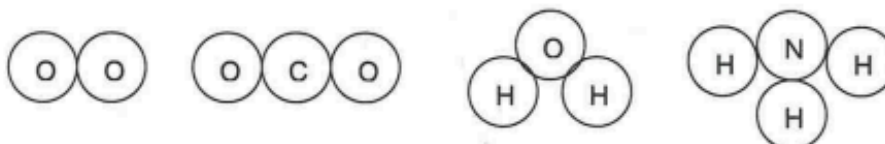
· The fill-in-the-blanks passage is an image: use the key words about particle arrangement, movement and energy (close together, random, vibrate, move faster, kinetic energy, forces, melting/boiling).

Questions about elements and compounds

- Which of the boxes contains:
 - Molecules of an element?
 - Molecules of a compound?
 - Atoms of an element?
 - Do any of the boxes contain a mixture? Explain your answer.



- Look at the following substances. Circle the elements and underline the compounds
 - Li
 - N_2
 - CO
 - Cl_2
 - S_8
 - NaCl
- How many elements make up each compound?
 - HCl
 - $CaCO_3$
 - HNO_3
 - H_2SO_4
 - CH_3OH
- How many atoms are there in each of these molecules?
 - N_2
 - CO_2
 - O_3
 - CH_4
 - SO_2
 - C_2H_4
- Circle the molecule with the most atoms, in each pair;
 - HCl or $POCl_3$
 - NH_3 or HNO_3
 - C_3H_8 or C_2H_5OH
 - CH_3COCH_3 or C_6H_6
- What is the formula for each molecules shown below



Define the term **mixture**?

Draw three particle diagrams to show the following:

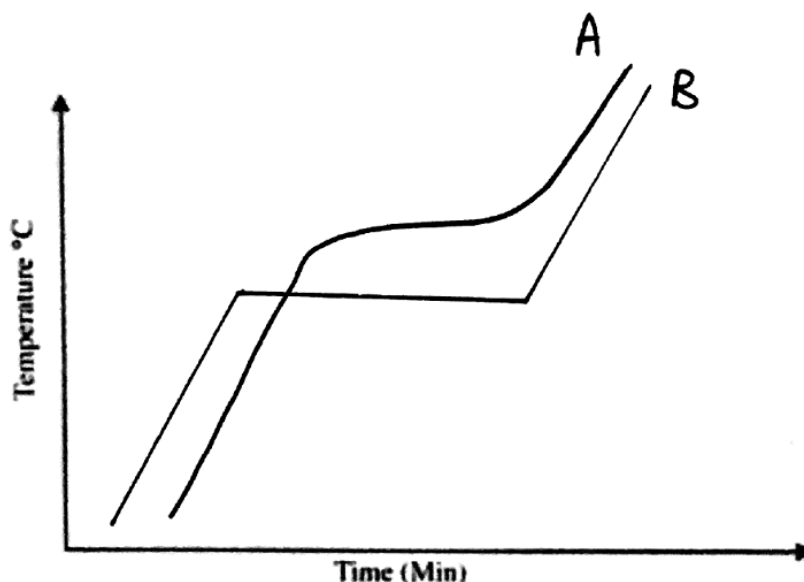
A solid mixture (e.g. a metal alloy) Draw two different types of particle mixed together but not joined to each other.	A 'liquid' mixture (e.g. seawater) Draw water particles with some dissolved solute particles mixed among them.	A gas mixture (e.g. air) Draw different gas molecules (e.g. nitrogen and oxygen) mixed randomly and far apart.
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Pure vs. Impure

1. In your own words, describe the difference between **pure** and **impure**.

A pure substance contains only one substance and has a sharp, fixed melting and boiling point. An impure substance has other substances mixed in, so it melts and boils over a range of temperatures.

2. The curves below were obtained by a student after heating two solid substances A & B.



- a) Which substance (A or B) has the higher melting point? ... A - its flat plateau (constant temperature) is at the higher temperature.

- b) Explain which curve represents an impure substance?
.....
.....

3. According to our definitions H_2O (water) is a compound. However, it is sometimes called a molecule. O_2 (oxygen gas) is also a molecule. Use this sentence to define a molecule.

A molecule consists of two or more atoms chemically joined together. These atoms can be of the same element or different elements.

Curve B is the impure substance: it changes state over a range of temperatures instead of at one sharp point, because impurities lower and spread out the melting point.

Extension: There are around 100 different elements. How many compounds do you think there are? (Hint: think about how many different combinations you could make with just 3 different elements.)

A huge number - millions. With about 100 elements there are almost endless combinations of two or more atoms, so countless compounds are possible.

**Blank page for notes about atoms, elements,
compounds and mixtures**

**Blank page for notes about atoms, elements,
compounds and mixtures**

Filtration & Crystallisation

Aim: To design and carry out practical work to try and separate pure samples of substances from a range of mixtures.

Safety: Goggles must be worn. Be careful using hot glassware. 

Design/Plan:

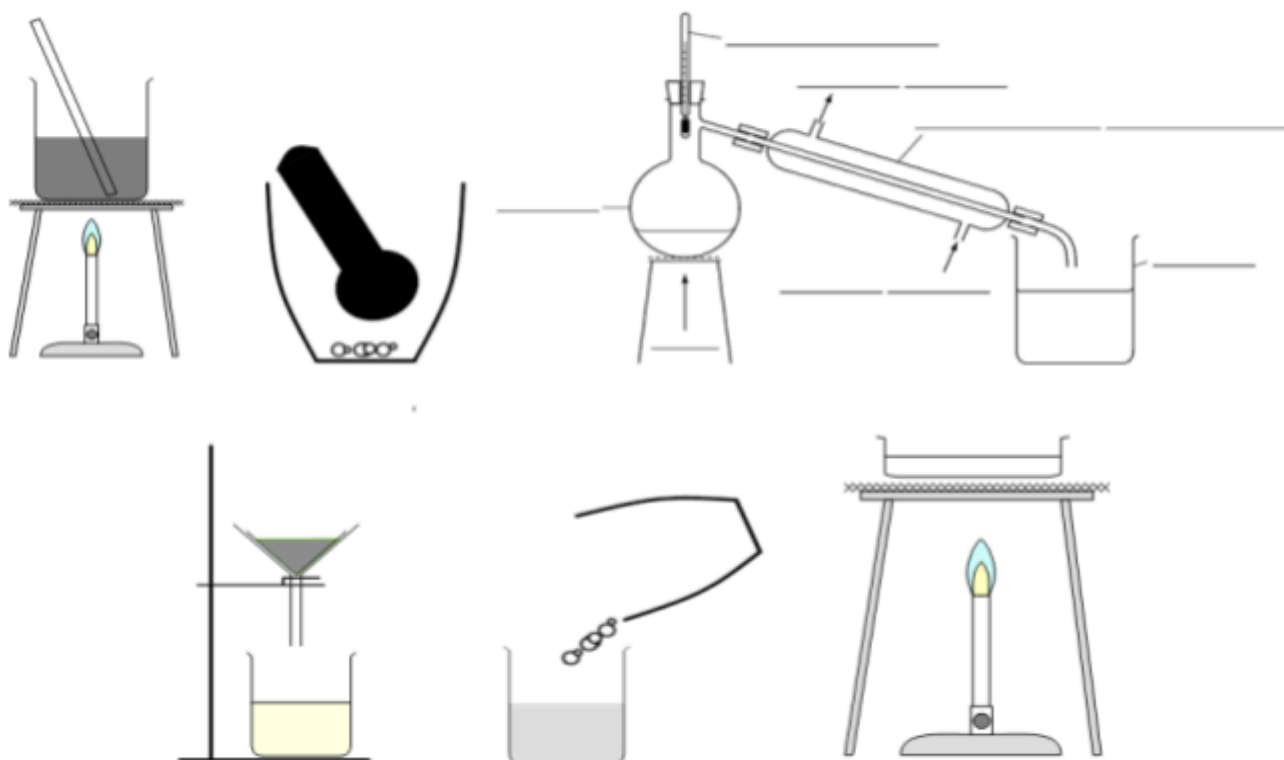
You have a practical problem to solve in your team of 2 or 3 students. You need to design an experiment to separate a pure sample of salt from the rock salt.

- Brainstorm your plan on the additional note space on the next page.
- Draw labelled diagrams showing how you will need to set up the apparatus and explain what happens at each stage.
- Write a method using numbered points to explain what you will do.

• Draw labelled diagrams of (1) filtering the rock salt mixture through filter paper in a funnel, and (2) heating the filtrate in an evaporating basin over a water bath.

Some of the apparatus below may help you decide what to do.

Diagrams and Apparatus:



1. Crush the rock salt and stir it into warm water so the salt dissolves. 2. Filter off the insoluble sand. 3. Heat the filtrate until crystals form, then cool and dry them.

Planning Notes - design an experiment to separate a pure sample of salt from the rock salt.

Analysis / Conclusion

1) Explain (in terms of the particles involved) how you were able to separate a pure sample of salt from the rock salt.

The salt dissolves and passes through the filter paper with the water, but the insoluble sand particles are too big to pass, so they are trapped. Heating evaporates the water, leaving salt crystals.

2) If you started with 5g of rock salt and weighed the mass of pure salt obtained at the end and it was 4g, what % of salt is there in rock salt? $4 / 5 \times 100 = 80 \%$, so rock salt is 80 % salt.

3) A student carried out this practical and the rock salt obtained was grey rather than pure white. What does this suggest and what could the student do next to test if the salt is pure?

It suggests the salt is impure (contains other substances). Test the melting point: pure salt melts sharply at one temperature, an impure sample melts over a range.

4) What is the name of the process called when the crystals of pure salt start to appear? If a student weighed their salt crystals before the water had all evaporated away, would this be an accurate mass of salt? What should they do to improve the accuracy of the mass?

crystallisation

No - the mass would be too high because not all the water has evaporated. Heat until all the water has gone (to constant mass), then weigh.

5) Henrique spilled some water onto a black kitchen worktop and did not wipe up the spill. The water looked clear. Eventually, the water dried but a white mark was left. What does this tell you about the water?

State water is pure: it has a fixed melting point/boiling point (0°C / 100°C) and its temperature stays constant while it changes

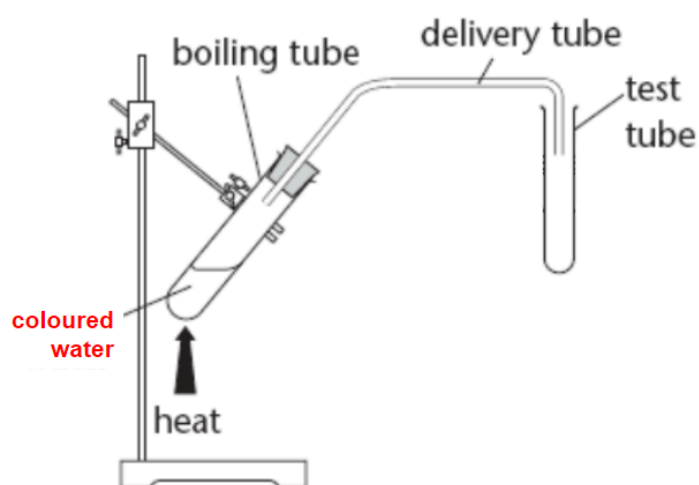
	True	False	Insufficient evidence
The water was mixed with a white substance	True (tick the True box)		
The water was pure		False (tick the False box)	
The water contained a dissolved substance		True (tick the True box)	
The water was salty			Insufficient evidence (tick the Insufficient evidence box)

Distillation

Aim: Carry out practical work to separate a pure sample of water from some inky water.

Safety: Goggles must be worn. Be careful using hot glassware. 

What is distillation? Distillation is a separation technique in which a liquid is boiled to form a vapour which is then cooled and condensed back into a pure liquid, separating it from a mixture.



What is the main issue with this apparatus?
The apparatus has no condenser to cool and condense the vapour, so the vapour escapes (and the thermometer bulb is in the liquid, not in the vapour).

Draw a labelled diagram of some better apparatus including a Liebig condenser.

· Draw: flask of inky water on a gauze/tripod being heated; still head leading into a Liebig condenser with water in at the bottom and out at the top; condensed pure water dripping into a beaker; thermometer bulb level with the side arm.

Analysis/conclusion:

1) Explain (in terms of the particles involved) how you were able to separate a pure sample water from your mixture.

The water particles gain enough energy to overcome the forces between them and become a gas, leaving the ink behind. The vapour is cooled in the condenser and condenses back to pure liquid water.

2) How could a student test the purity of the water they obtained from this experiment?

Measure its boiling point - pure water boils at exactly 100 °C; impure water boils over a range of temperatures.

3) You can buy 'distilled' water from many shops. What does this tell us about the purity of the water that we are buying? It has been purified by distillation, so it contains very few dissolved substances - it is (almost) pure water.

4) Why do we buy 'mineral' water instead of 'distilled' water? What are the advantages of each?

Mineral water contains dissolved minerals the body needs and tastes better; distilled water is pure with no dissolved substances, so it is used in laboratories, car batteries and irons (no limescale).

5) How is fractional distillation is different from simple distillation.

Fractional distillation uses a fractionating column and separates a mixture of liquids with different boiling points; simple distillation separates one liquid from a solution.

6) A mixture of three liquids needs separating. The expected temperature range for collecting each liquid in the distillate is shown below.

liquid X: 40°C-100°C

liquid Y: 120°C-160°C

liquid Z: 180°C -200°C

The mixture was added to the distillation apparatus and was heated quickly to 150°C. Select the row of the table that shows what would be collected in the distillate.

Liquid X and liquid Y (the row showing X and Y collected, not Z)

	liquid X	liquid Y	liquid Z
A	✓	✓	✓
B	✓	X	X
C	✓	✓	X
D	X	✓	X

· Select the table row showing that both X and Y are collected: heating to 150 °C boils off X (40-100 °C) and Y (120-160 °C); Z (180-200 °C) is not collected.

Chromatography

Aim: Describe these experimental techniques for the separation of mixtures using paper chromatography. Understand how a chromatogram provides information about the composition of a mixture and how to use the calculation of R_f values to identify the components of a mixture.

Apparatus: Strips of chromatography paper, beakers, felt tip pens (dark colours, brown, black, purple green). Plus an unknown black pen of a different brand.

Method:

1. Draw a pencil line about 1.5 cm from the bottom of the strip of chromatography paper.
2. Put a small mark of the felt tip on the line on this line of the chromatography paper.
3. Repeat with other colours of felt tip about 2 - 3 cm from the previous spot.
4. Label each spot in pencil underneath the line.
5. Hang the chromatography paper using a splint in a beaker. (If needed, you can use a paperclip to hold it in place.) Put the paper in a beaker containing 1 cm depth of water, making sure the pencil line is about 0.5 cm above the level of the water and the paper doesn't touch the sides of the beaker.
6. Allow the water (solvent) to rise up the paper. The greater the time for separation the more accurate the results. This should take about 10-15 mins but you need to keep checking it.
7. Remove the paper before the solvent reaches the top, mark the solvent (water) line (solvent front) with a pencil.
8. Allow the paper to dry and record the distances that the different coloured dyes have moved with a ruler measuring in mm and put this into the table.
9. Determine the retention factors for each of the coloured dyes obtained using the following equation:

$$\text{Retention factor } (R_f) = \frac{\text{distance moved by solute (dye)}}{\text{distance moved by the solvent (water)}}$$

10. On a separate piece of chromatography paper, set up another chromatogram for the unknown ink in the pen given. Determine the retention factor of any dyes in this ink sample.

Results: Distance moved by the solvent (water) = 80 mm

Colour	Distance (mm)	Retention factor (R_f)
blue	75	0.94
red	60	0.75
yellow	48	0.60
green	30	0.38

Unknown ink:

Distance moved by the solvent (water) = 80 mm

Determine the retention factors: $R_f = \frac{\text{distance moved by the dye}}{\text{distance moved by the solvent, e.g. } 75/80 = 0.94}$

Analysis/Conclusion:

1. Which colour of felt tip contained the most different dyes?

Black - it separated into the largest number of different coloured spots.

2. Do any colours contain only one dye? If the ink was made of only one pure dye, how many spots would you expect to see? Yes - some colours gave only one spot. An ink made of only one pure dye would give just one spot.

3. Did any of the ink from the felt tips not move up the filter paper? How could this happen?

Yes - any dye that is insoluble in water does not dissolve, so it stays on the pencil starting line and does not move up the paper.

4. What would happen if the level of the water (solvent) was higher than the ink at the start of the experiment? The spots would dissolve into the solvent and be washed off the paper, so the dyes would not separate properly.

5. Did the unknown felt tip pen contain any of the same dyes in the ink as the other samples? How can you tell?

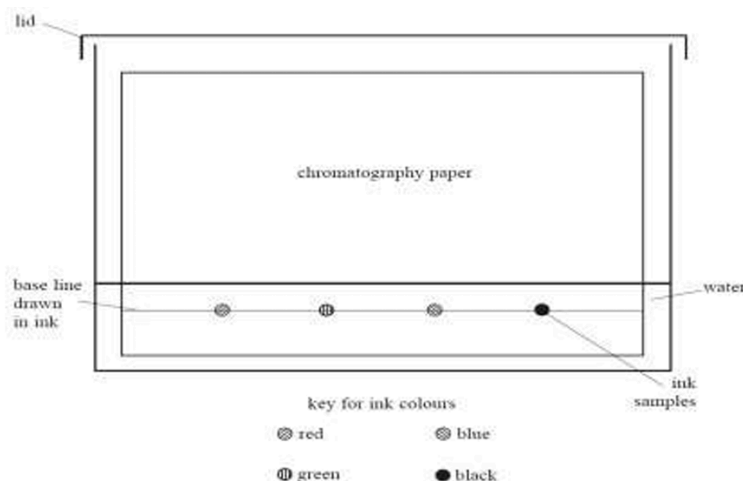
Yes, if a spot from the unknown pen travelled the same distance (same R_f value) as a dye from one of the other samples, it must be the same dye.

Extension: How could you improve the accuracy of this experiment? Explain why some dyes move greater distances on the paper than others.

Let the solvent run for longer and measure the distances more precisely. Dyes that are more soluble in the solvent and less strongly attracted to the paper travel further.

Chromatography Exam Questions

1. A student investigates the colours contained in inks from felt-tip pens. He uses chromatography and sets up his experiment as shown:



(a) Identify two mistakes in the way he sets up the experiment. For each mistake, state what problem it would cause.

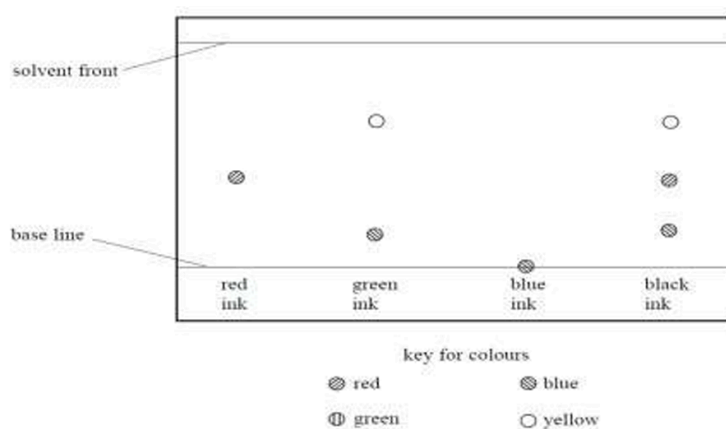
1st mistake = the spots/starting line are below the surface of the solvent (water)

Problem = the inks would dissolve into the solvent instead of travelling up the paper
so the dyes would not separate (no chromatogram would form)

2nd mistake = the beaker has no lid (and the paper touches the side of the beaker)

Problem = the solvent would evaporate, so the solvent front would not rise properly
and the results would be uneven and inaccurate

(b) Another student repeats the experiment, but does not make any mistakes. She uses inks from four different felt tip pens. The diagram shows her results.



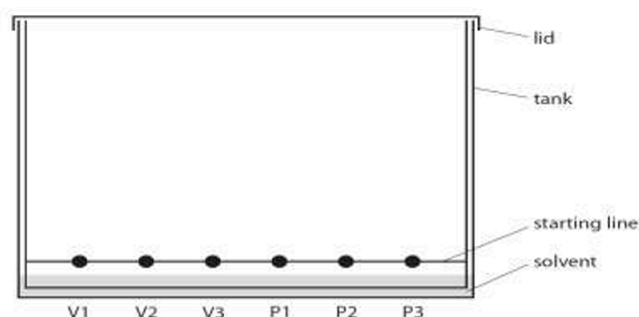
(c) (i) How many different colours does the black ink contain? count the number of separate spots produced by the black ink, e.g. 4 colours

(ii) Which of the inks tested could be mixed together to make the black ink? the inks whose spots are at the same height (same R_f value)

(iii) Which of the inks tested is insoluble in water? Explain your answer. _____

because it is insoluble in water, so it does not dissolve and cannot be carried up the paper by the solvent

2. A student investigates the pigments found in some vegetables and fruit. She obtains some coloured vegetable and fruit extracts from carrots, tomatoes and sweet potatoes. She places a spot of each extract on chromatography paper, along with spots of the three pigments beta-carotene, chlorophyll and lycopene. Her teacher provides a solvent containing volatile, flammable, organic compounds for the experiment. The diagram shows the apparatus at the start of the experiment:



Key to vegetable and fruit extracts and pigments

V1 = carrots V2 = tomatoes V3 = sweet potatoes

P1 = beta-carotene P2 = chlorophyll P3 = lycopene

She separates the pigments by paper chromatography. The spots show which pigments are present and their R_f values identify them (e.g. chlorophylls, carotenes) against known values.

(a) Explain why it is important for the solvent level to be below the spots. _____

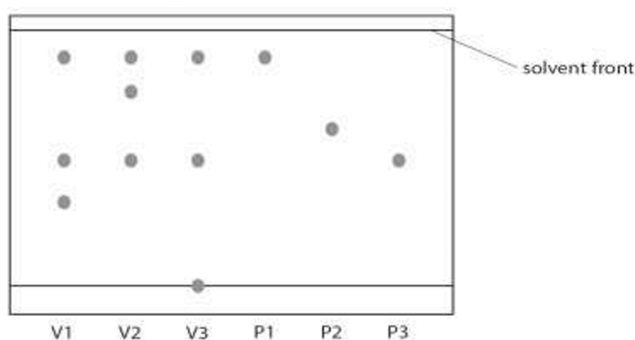
so the pigments would not separate and no chromatogram would form

(b) State two potential problems that are prevented by fitting the tank with a lid. _____

the solvent vapour escaping, so the solvent front rises unevenly/inaccurately

and the results are not repeatable between runs

The diagram shows the chromatogram at the end of the experiment.



(c) Put a tick next to **three** statements that are supported by the chromatogram.

- | | |
|-------------------------------------|---|
| ☒ | Chlorophyll is not present in carrots, sweet potatoes or tomatoes |
| ☒ | Beta-carotene is present in carrots but not present in tomatoes. |
| ☐ | Both beta-carotene and lycopene are present in sweet potatoes. |
| ☒ | Lycopene is present in tomatoes but not present in carrots. |
| ☐ | Both carrots and tomatoes contain a pigment other than beta-carotene, chlorophyll and lycopene. |

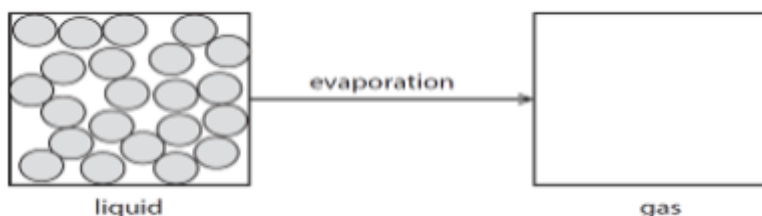
(d) Suggest a reason why there is a spot on the starting line in the chromatogram for sweet potatoes.

some of the pigments in sweet potato are insoluble in the solvent (or hold onto the paper too strongly), so they do not move up the paper

Blank page for notes about separating techniques

Questions on Matter and Separating Techniques

1. When a liquid evaporates at room temperature, it changes into a gas.
The diagram shows the arrangement of the particles in a liquid.



(a) Complete the diagram to show the arrangement of four particles in a gas. (1)

(b) Describe the movement of particles in a gas. (1)

· Draw four particles widely spaced apart and randomly arranged in the box (not touching).

(c) Explain why heating a liquid causes it to evaporate more quickly. (2)

They move quickly and randomly in all directions,
colliding with each other and the walls of the container.

2. Techniques used in the separation of mixtures include

A = crystallisation

B = filtration

C = fractional distillation

D = simple distillation

For each separation, select the most suitable technique, A, B, C or D, used to obtain the first named substance from the mixture. Each letter may be used once, more than once or not at all.

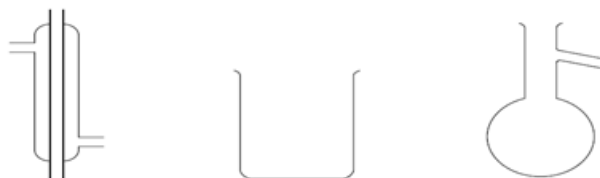
(a) Pure water from sea water = D (simple distillation)

(b) Ethanol from a mixture of ethanol and water = C (fractional distillation)

(c) Calcium carbonate from a mixture of calcium carbonate and water = B (filtration)

(d) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s})$ from $\text{CuSO}_4(\text{aq})$ = A (crystallisation) (4)

(e) Sea water contains dissolved sodium chloride. The following pieces of laboratory apparatus can be used to make drinking water from sea water.



(i) Draw a labelled diagram to show how these pieces of apparatus can be assembled to do this. (3)

Heating gives the liquid particles more kinetic energy, so more particles have enough energy to overcome the forces holding them in the liquid and escape as a gas.

· Draw the heated flask of sea water connected by a delivery tube/Liebig condenser into a collecting test tube or beaker, cooled so the water vapour condenses.

(ii) Name the technique used in this process. (1)

simple distillation

3. A teacher uses this apparatus to show the reaction between ammonia gas and hydrogen chloride gas to form ammonium chloride.



Concentrated ammonia solution gives off ammonia gas. Concentrated hydrochloric acid gives off hydrogen chloride gas. The reaction between the gases can be represented by this equation.



The diagram below shows the apparatus after a few minutes.



(a) Which process occurs when the gas particles move along the tube? (1)

(b) Name the white solid that forms in the tube. (1)

(c) Explain how the diagram shows which gas moves more quickly during the experiment. (1)

diffusion

ammonium
chloride

(d) The particles in a gas move very quickly. Why does it take a few minutes for the white gas to form? (1)

The white ring forms nearer the hydrogen chloride end, so the ammonia particles travelled further in the same time - ammonia moves faster.

The particles move randomly and collide with the air particles, so they take time to diffuse to the middle of the tube and react.

4. This is a description of how the orange colouring can be extracted from rose petals.

- crush the petals using a pestle and mortar
- add the crushed petals to some ethanol in a beaker
- heat to about 60°C and stir to produce an orange solution
- separate the orange solution from the petals

(a) (i) Suggest why ethanol is used instead of water. (1)

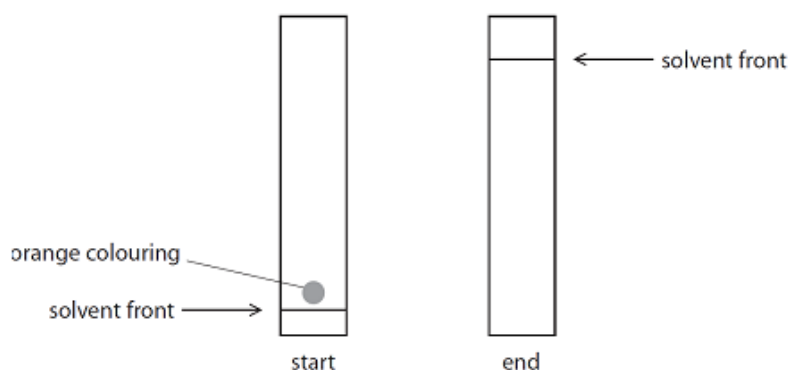
The orange colouring
is more soluble in
ethanol than in water,
so it dissolves out of
the petals more easily.

(ii) Ethanol is a flammable liquid. Suggest how it could be heated safely. (1)

(iii) How could the orange solution be separated from the petals? (1)

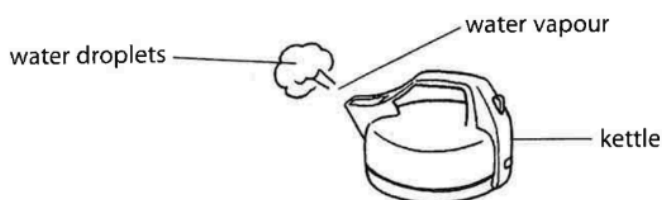
By filtration (the solution passes through the filter paper, leaving the petals behind).

(b) The orange colouring is analysed using chromatography and is found to consist of two different colours, red and yellow. The diagram shows the chromatography paper at the start of the experiment. Complete the diagram to show a possible result at the end of the experiment. (2)



5.

The diagram shows a kettle of boiling water.



As the water vapour cools it turns into droplets of liquid water.

(a) The change of state when water vapour changes into liquid water is described as

(1)

- ☐ A boiling
- ☐ B condensation
- ☐ C evaporation
- ☐ D sublimation

(b) Describe what happens when water vapour cools to form liquid water.

Your answer should include the change in the energy, arrangement and movement of the particles.

(3)

change in energy

.....

.....

change in arrangement

.....

.....

change in movement

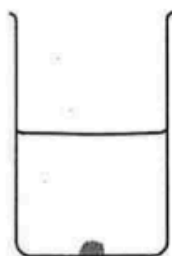
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6.

Hydrated copper(II) sulfate is a soluble blue solid. A large crystal of this solid is placed at the bottom of a beaker of water.

The diagram shows the beaker immediately after placing the crystal in it, and after two days.



after placing the crystal



after two days

- (a) After two days, the crystal becomes smaller and the liquid near the bottom of the beaker becomes blue.

Which statement explains these observations?

(1)

- | | |
|--|---|
| ☐ A the crystal dissolves | ☐ C the crystal melts |
| ☐ B the crystal freezes | ☐ D the crystal sublimates |

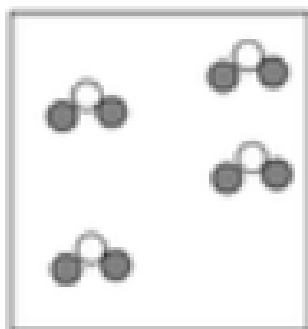
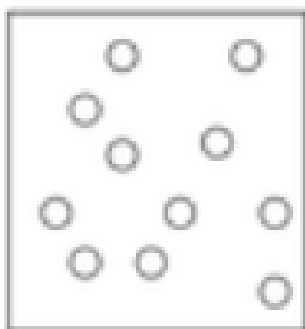
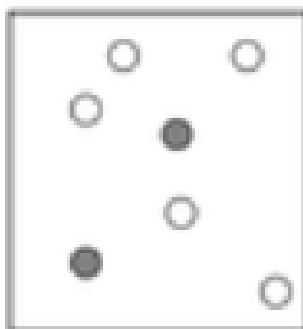
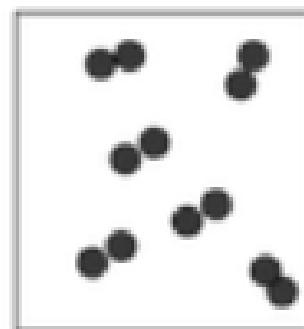
- (b) After two weeks, the crystal has disappeared.

Which statement best describes the appearance of the liquid in the beaker after two weeks?

(1)

- | | |
|--|--|
| ☐ A it is all blue | ☐ C only the lower part is blue |
| ☐ B it is all brown | ☐ D only the upper part is blue |

- (c) Describe each of the following boxes as either 'element', 'compound' or 'mixture'.

elementcompoundmixtureelement

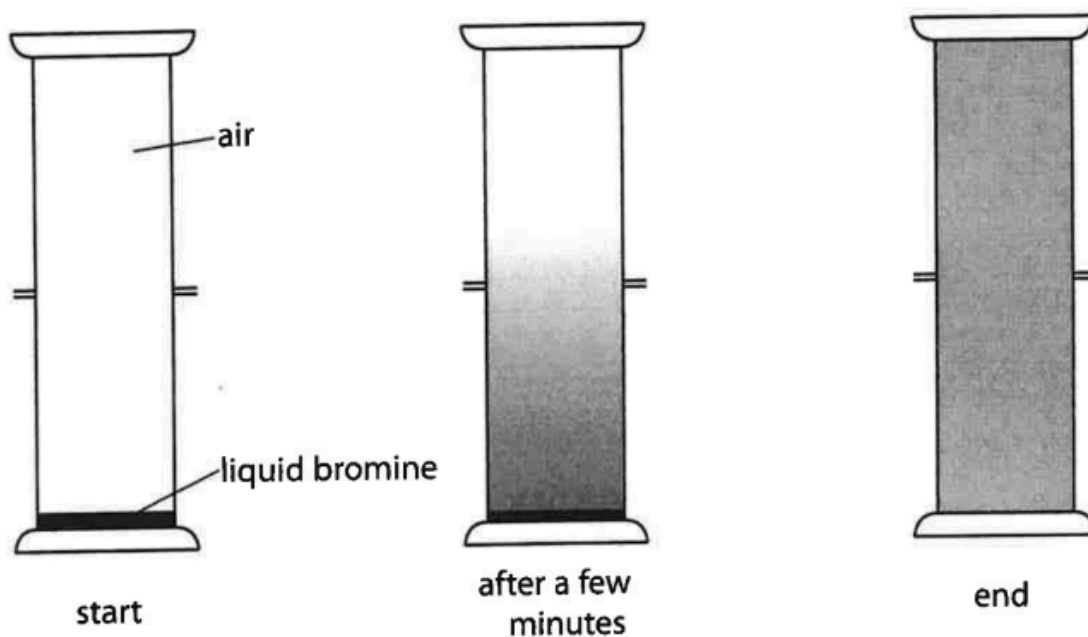
Question 6 is largely image-based - the text for parts (a) and (b) was not extracted, so only part (c) can be answered.

7.

A teacher demonstrates, in a fume cupboard, two experiments to show the movement of particles.

- (a) In the first experiment she places some liquid bromine at the bottom of a gas jar. She then places another gas jar containing air on top of it, as shown in the diagram.

The diagram shows the apparatus at the start, after a few minutes and at the end of the experiment.



Place crosses (☒) in **two** boxes to show which statements are correct about this experiment.

- ☐ **A** All the air particles in the upper gas jar stay there.
- ☐ **B** Bromine and air react to form bromine oxide.
- ☐ **C** Bromine has a darker colour than air.
- ☐ **D** Bromine vapour diffuses upwards.
- ☐ **E** Liquid bromine sublimates during the experiment.
- ☐ **F** The concentration of bromine in the lower gas jar does not change.

· Question 7 is an image-only question - no text was extracted for it, so no answer can be given from this file.

8.

(a) The list shows some techniques used to separate mixtures.

- A** crystallisation
B filtration
C fractional distillation
D paper chromatography
E simple distillation

Complete the table to show the best method of obtaining each substance from the mixture.

In each case, choose one of the letters A, B, C, D or E. Each letter may be used once, more than once or not at all.

(2)

Substance	Mixture	Letter
sand	sand and water	
solid copper(II) sulfate	aqueous copper(II) sulfate	

(b) Gold occurs in ores, which are mixtures of gold and other substances. Several elements and compounds are used in the extraction of gold from its ores.

Each box below represents the substances present in one part of the extraction process.

Classify the contents of each box as a compound, an element or a mixture by writing your choice below each box.

(1)

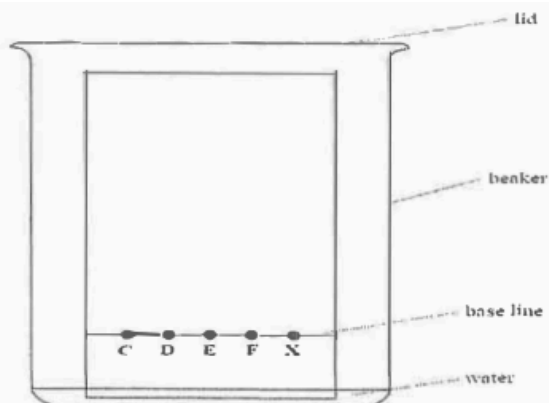
Compound, element or mixture			

Q8 is printed entirely as figures (image blocks y93-547 and y551-746). The text layer contains no question lines, blanks or cells for this page, so the Q8 answers must be completed on the page image itself.

Complete each labelled prompt inside the two Q8 diagrams (e.g. apparatus/table parts and their written answers) directly in the spaces printed inside the figures - extraction gap, verify against the page image.

9.

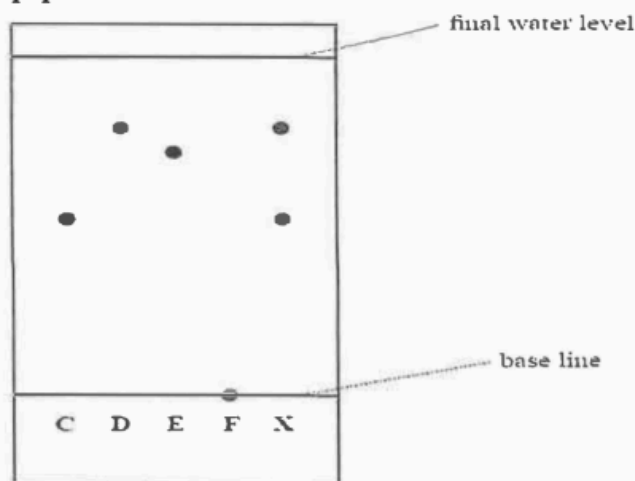
Four Separate food dyes (C, D, E and F) and a mixture of food dyes (X) were investigated using paper chromatography. The diagram shows the apparatus used:



(a) Why should the water level be below the food dyes? (1)

(b) During the experiment the water rises up the paper. The experiment is stopped just before the water reaches the top of the paper.

The diagram shows the paper after it has been removed from the beaker and dried.



(i) Which of the food dyes C, D, E and F does X contain? (1)

(ii) Suggest why food dye F did not move up the paper during the experiment. (1)

(c) Each food dye has an R_f value.

Record the distances for food dye D in the table below and calculate its R_f value.

(3)

The Q9 apparatus/instructions and any question parts are printed inside the figure images (y122-368 and y372-690) and are not present in the text layer: read them from the page image

Distance moved by food dye D from base line in mm	45
Distance moved by solvent front from base line in mm	90
R_f value $R_f = \text{distance moved by dye D} / \text{distance moved by solvent front} = 45 / 90$	0.50 (no units)

Blank page for notes about anything from Unit 1

~~· Blank page for notes on Unit 1 - no set answers required; students' own revision notes.~~

Appendix - answers with no space on the original page

These belong to the printed question on the page shown.

p2: Element: only one type of atom. Compound: two or more elements chemically combined in a fixed ratio. Mixture: substances not chemically combined, separable by physical means.

p2: A pure substance has a sharp, fixed melting point and boiling point. Impurities lower the melting point, raise the boiling point and widen the melting/boiling range.

p2: An atom is the smallest particle of an element that can exist. A molecule is a group of two or more atoms chemically (covalently) bonded together.

p2: An atom is the smallest particle of an element that can exist. A molecule is a group of two or more atoms chemically (covalently) bonded together.

p2: Crystallisation: evaporate some solvent, then cool to form crystals. Simple distillation: boil the solution and condense the vapour to collect the solvent (pure water from salt solution).

p2: A chromatogram shows how many components (spots) a mixture contains; each spot is a different substance, and its position/R_f value can identify it alongside known substances.

p2: $R_f = \text{distance moved by the spot} \div \text{distance moved by the solvent front}$. R_f is constant for a substance under the same conditions (solvent, paper, temperature).

p2: Separate the dyes in inks/food colourings by paper chromatography: pencil baseline, small spots, solvent (water) below the spots; compare spots and R_f values with known colourings.

p9: At the same temperature all gas particles have the same average kinetic energy ($KE = \frac{1}{2}mv^2$). A smaller mass must therefore have a greater speed, so lighter gases diffuse faster.

p27: A white ring (smoke) of ammonium chloride forms where the two gases meet. It forms nearer the hydrogen chloride end because ammonia (M_r 17) diffuses faster than hydrogen chloride (M_r 36.5).

p27: Heat it in a water bath (or with an electric hot plate), not with a naked flame.

p28: Question 5 is an image-only question - no text was extracted for it, so no answer can be given from this file.